

The stick number of 8_1 : a computer-assisted proof with a partially formalized geometric reduction

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Draft of September 10, 2026

Abstract

We present a computer-assisted argument that the Rolfsen knot 8_1 has stick number ten. An exposed-vertex projective construction associates to every hypothetical nine-stick representative a diagram with eight shadow edges, one of which passes over every crossing. Necessary determinant-sign, height-compatibility, blocked-ear, and diagram-invariant conditions give a Boolean formula on 142 geometric signs. Its explicit gate encoding has 18,597 variables and 76,298 clauses and is refuted by a fixed LRAT certificate. Lean 4 verifies this fixed refutation, the source-to-CNF translation, and the real-geometric validity of 2,283 of the 2,294 source constraints. The remaining nine ear conditions and two invariant conditions, as well as the global topological reduction, are justified in ordinary mathematics and are explicitly outside the formalized theorem. An exact rational ten-gon and a checked Reidemeister trace provide a separate upper bound. We describe the certificate interfaces and trust boundary so that the computation and its geometric interpretation can be reviewed separately.

1 Introduction

The stick number $\text{stick}(K)$ of a tame knot K is the least number of straight segments in an embedded polygon representing K in $\mathbb{R}^3$. Lengths are unrestricted; this is not the equilateral stick number. Mirroring a polygon preserves its number of edges.

Theorem 1.1. *The Rolfsen knot 8_1 satisfies $\text{stick}(8_1) = 10$.*

Calvo’s classification of octagonal knots [1, Theorem 1(iv)] excludes 8_1 and its mirror, and hence gives $\text{stick}(8_1) \geq 9$. Subdividing edges explains why the same classification excludes every smaller polygon. The construction of Huh, No, and Oh [2, Theorem 1.1] gives $\text{stick}(K) \leq c(K) + 2$ for a two-bridge knot of crossing number at least six. It therefore gives the upper bound ten for 8_1 . We also supply an independent exact upper-bound certificate. Neither external theorem supplies the nine-stick exclusion proved here. We make no claim that the present work is a classification of all nine-stick knots.

The lower-bound argument uses only necessary conditions. In particular, we never assert that every satisfying abstract sign assignment is realizable by points in $\mathbb{R}^3$. An unsatisfiable relaxation suffices: a geometric counterexample would induce a satisfying assignment, whereas the fixed formula admits none. The eleven source conditions not yet derived from geometry inside Lean are mathematical obligations, not omitted clauses of the finite formula. All 2,294 constraints at the source level participate in the certified exclusion.

The argument has three distinct layers:

1. Ordinary geometry and knot theory put any counterexample into a positive, nondegenerate projective frame and identify its diagram invariants.
2. Real-algebraic Lean proofs establish the determinant and order conditions of this frame. A conditional formal theorem exposes precisely the eleven remaining source-root premises.
3. A formal source-to-circuit-to-CNF translation and an accepted LRAT refutation exclude the finite conjunction.

The conclusion is a computer-assisted theorem with partial formalization, not an end-to-end Lean theorem about ambient-isotopy classes of knots.

2 The exposed-vertex construction

Write a hypothetical embedded nine-gon in cyclic order as $p_0, \dots, p_7, v$. Embeddedness and knot type are stable under sufficiently small perturbations of its vertices. Here is an explicit ambient extension which avoids any assumption about extending arbitrary topological isotopies. Triangulate a large cube compatibly with the polygon, allowing subdivision of its edges. Prescribe a displacement at each original polygon vertex, interpolate it linearly along polygon edges, set it to zero at all other triangulation vertices, and extend affinely on each simplex. The resulting continuous displacement u vanishes on the cube boundary; extend it by zero outside. Since the triangulation is finite, sufficiently small vertex displacements make the global Lipschitz constant L of u less than one. Then $H_t(x) = x + tu(x)$ is an ambient isotopy: injectivity follows from

$$|H_t(x) - H_t(y)| \geq (1 - tL)|x - y|,$$

and surjectivity follows by applying the contraction principle to $x \mapsto y - tu(x)$ for each y . The inverses are jointly continuous by the same uniform estimate. Each polygon edge is mapped linearly onto the corresponding perturbed edge. This proves both embeddedness and knot preservation for sufficiently small vertex perturbations. This finite triangulation argument is ordinary mathematics, not a Lean theorem here.

Choose a vertex v exposed by a supporting linear functional. After an affine coordinate change, $v = 0$ and $p_i = (x_i, y_i, z_i)$ with $z_i > 0$. Put

$$q_i = (x_i/z_i, y_i/z_i), \quad h_i = 1/z_i, \quad P_i = (q_i, h_i).$$

Here and below shadow indices are taken modulo eight.

Lemma 2.1 (Roof diagram). *The polygon has a diagram, up to mirror image, consisting of the projected chain $q_0q_1 \cdots q_7$ closed by q_7q_0 . Crossings among chain edges use the affine heights h_i . At every crossing involving the closing shadow edge, that edge is the overpassing strand. The closing edge represents two spatial segments.*

Proof. On the open half-space $z > 0$ the projective map $F(x, y, z) = (x/z, y/z, 1/z)$ is a diffeomorphism and maps segments to segments. To check its effect on knot type, set

$$d_\theta = \cos \theta + z \sin \theta, \quad G_\theta(x, y, z) = \left(\frac{x}{d_\theta}, \frac{y}{d_\theta}, \frac{z \cos \theta - \sin \theta}{d_\theta} \right), \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

The denominator is positive and the projective transformation is invertible on its image. Segments remain segments, so for a compact polygon in this half-space these maps give a continuous path of embedded polygons from the identity to F followed by reflection in the

last coordinate. Along this path the knot type is locally constant by the small-perturbation construction above; compactness of the parameter interval gives a finite concatenation of its ambient isotopies. This justifies “up to mirror image” without inferring knot equivalence merely from an arbitrary diffeomorphism of an open set.

Choose r in the interior of q_7q_0 , away from all its crossings. Replace v by $\varepsilon(r_x, r_y, 1)$ with $\varepsilon > 0$ sufficiently small. Stability preserves the knot type. Its image under F is $(r, 1/\varepsilon)$. The two incident projected segments subdivide q_7q_0 . At each of its finitely many crossings with the chain, the coefficient of $1/\varepsilon$ in the interpolated height is strictly positive. Thus all these heights eventually exceed the chain height. The two pieces have the same oriented shadow direction, so the subdivision introduces no diagram crossing. $\square$

We choose the initial polygon generically so that every determinant used below is nonzero: all four-point spatial volumes, all three-point shadow orientations, and all three-line shadow brackets. These conditions are compatible with the exposed vertex and with knot preservation. The coordinates (q_i, h_i) vary in an open set with $h_i > 0$. Each prohibited determinant is a nonzero polynomial in these coordinates. For spatial volumes and planar orientations this follows by freely moving one of the involved points. Three pairwise nonadjacent shadow lines can be chosen independently, so their concurrency determinant is not identically zero. If an adjacent pair occurs, the factorization in Section 3 reduces the assertion to nonzero orientation polynomials. A finite union of proper polynomial zero sets has empty interior. It can therefore be avoided in the open set of admissible perturbations. In particular, no exceptional coordinate family is silently excluded from the universal nine-gon argument.

3 Signs and necessary algebraic constraints

Use homogeneous columns

$$w_i = (q_{ix}, q_{iy}, h_i, 1) \quad (0 \leq i < 8), \quad w_8 = (0, 0, 1, 0).$$

Our alternating sign convention is $\chi(i, j, k, l) = \text{sgn}(-\det(w_i, w_j, w_k, w_l))$. For finite indices this is the sign of

$$V(i, j, k, l) = \det(P_j - P_i, P_k - P_i, P_l - P_i),$$

and $\chi(i, j, k, 8) = \text{sgn } T(i, j, k)$, where $T(i, j, k) = \det(q_j - q_i, q_k - q_i)$. Reflecting the first planar coordinate if necessary makes $T(0, 1, 2) > 0$. The corresponding mirror has the same crossing number and a_2 .

There are $\binom{9}{4} = 126$ independent sorted point signs. For shadow edge i set $u_i = q_{i+1} - q_i$ and

$$L_i = (-u_{iy}, u_{ix}, \det(q_i, u_i)), \quad C(i, j, k) = \det(L_i, L_j, L_k).$$

Here $L_i \cdot (x, y, 1) = 0$ is its supporting line. Of the triples of shadow edges, sixteen have no adjacent pair. Their bracket signs are additional Boolean inputs. If $b = a + 1$ modulo eight, direct expansion gives

$$C(i, a, b) = T(i, i + 1, b)T(a, b, b + 1). \quad (1)$$

Alternation supplies every other order of the indices. Thus only 142 geometric Boolean inputs are needed. Positive signs are encoded as true; negation changes the sign, and Boolean equality multiplies two signs.

The three-term Grassmann–Pluecker identities imply that their three signed nonzero products cannot all have the same sign. Apply these identities to rank-four point brackets and rank-three line brackets. There are respectively 1,260 and 280 resulting conditions. The relation

$$C(ab, cd, ef) = T(a, b, e)T(c, d, f) - T(a, b, f)T(c, d, e) \quad (2)$$

gives 48 additional mixed-sign conditions. Consistency of the adjacent factorization gives another 48.

The columns w_i have a common strictly positive functional: their third coordinate is $h_i > 0$ for finite i and one for $i = 8$. For any five columns, their alternating four-minor dependence has coefficients of both signs; otherwise applying this functional gives a nonzero quantity of one sign equal to zero. The analogous statement for $(q_{ix}, q_{iy}, 1)$ uses the last-coordinate functional. These give 126 homogeneous and 70 projected acyclicity conditions. Normalization adds one condition. All these are necessary properties of real points; no converse realizability theorem is used.

4 Height compatibility

Only edges $0, \dots, 6$ are finite spatial lines. For each choose an affine function $F_i(x, y)$ taking the required endpoint heights, and write F_i also for its coefficient vector. Such a function exists because projected endpoints are distinct. Define

$$H_{ij} = V(i, i+1, j, j+1) = (F_i - F_j) \cdot (L_i \times L_j).$$

The second equality is a determinant identity. It is independent of the arbitrary extension of the height function away from its line. For adjacent indices $H_{ij} = 0$; for nonadjacent finite edges it is nonzero.

Lemma 4.1 (Height stresses). *If $\sum_i \lambda_i L_i = 0$, then*

$$\sum_{i < j} \lambda_i \lambda_j H_{ij} = 0. \quad (3)$$

More generally, if a symmetric matrix Ω satisfies $\sum_j \Omega_{ij} L_j = 0$ for every i , then $\sum_{i < j} \Omega_{ij} H_{ij} = 0$.

Proof. In the latter sum the coefficient paired with F_i is $\sum_{j \neq i} \Omega_{ij} (L_i \times L_j) = L_i \times \sum_j \Omega_{ij} L_j = 0$. Taking $\Omega_{ij} = \lambda_i \lambda_j$ proves the first assertion. $\square$

For each four-subset of the seven finite lines, use the alternating three-minor dependence for λ . Remove terms with adjacent edges and translate the remaining zero sum into the requirement that its nonzero terms have both signs. This supplies $\binom{7}{4} = 35$ conditions.

A useful strengthening comes from five lines in order (h, a, b, c, d) . Let the dependence on (h, a, b, c) be

$$v = (C(a, b, c), -C(h, b, c), C(h, a, c), -C(h, a, b), 0)$$

and the dependence on (h, a, b, d) be

$$w = (C(a, b, d), -C(h, b, d), C(h, a, d), 0, -C(h, a, b)).$$

Define $\Omega = w_a w_b v v^\top - v_a v_b w w^\top$. It satisfies the lemma; its off-diagonal entries at ab and cd vanish. The remaining eight coefficients factor as

$$\begin{aligned} \Omega_{ha} &= -v_a w_a C(h, a, b) C(a, c, d), \\ \Omega_{hb} &= v_b w_b C(h, a, b) C(b, c, d), \\ \Omega_{ic} &= w_a w_b v_i v_c \quad (i = h, a, b), \\ \Omega_{id} &= -v_a v_b w_i w_d \quad (i = h, a, b). \end{aligned}$$

These factorizations follow from the same three-term bracket identity and are also proved over $\mathbb{R}$ in Lean. Their factors are all nonzero. Deleting adjacent-edge terms again leaves a

mixed-sign zero sum. For each five-subset choose one of five hubs and one of three pairings of the other four lines. The resulting $\binom{7}{5} \cdot 5 \cdot 3 = 315$ conditions are the five-line height constraints.

None of these height stresses is imposed on the overpassing closing shadow edge. Giving that edge an affine height would change the class of spatial polygons and invalidate the reduction.

5 The nine blocked-ear conditions

For an embedded polygon in general position, an ear at a vertex is the closed triangle spanned by that vertex and its two neighbours.

Lemma 5.1. *Every ear of a nine-stick 8_1 is pierced through its relative interior by an edge whose endpoints are not vertices of that triangle.*

Proof. If no such edge pierces the triangle, slide the two incident sides across its disk onto the third side and remove the redundant vertex. More explicitly, move the middle vertex linearly toward an interior point of the third side. This gives a path of embedded nine-gons ending at a subdivision of the eight-gon; the small-perturbation argument of Section 2, applied along the parameter interval, preserves ambient knot type. General position rules out boundary intersections: a nonincident edge meeting a triangle side would make four vertices coplanar, and an edge incident to a triangle vertex leaves its plane immediately. The disk's interior is otherwise disjoint from the knot by hypothesis. The resulting eight-edge polygon represents the same knot, contrary to Calvo's classification. Of the nine polygon edges, exactly five avoid all three vertices of a given ear, so these are precisely the eligible edges. $\square$

For distinct vertices a, b, c, s, t the piercing condition is

$$V(a, b, c, s)V(a, b, c, t) < 0, \tag{4}$$

$$V(s, t, a, b)V(s, t, b, c) > 0, \tag{5}$$

$$V(s, t, a, b)V(s, t, c, a) > 0. \tag{6}$$

To verify it, intersect the segment's line with the triangle plane. Equation (4) places the unique intersection strictly between s and t . Cramer's rule expresses its three barycentric coordinates as a common nonzero factor times the three quantities in (5)–(6). They are all strictly positive after normalization exactly when those quantities have the same sign. This proves both directions, including the strict-interior requirement.

One must also justify the use of index eight, which denotes an ideal point rather than a finite vertex. Before the apex perturbation in Lemma 2.1, apply the invertible homogeneous linear map

$$(x, y, z, t) \mapsto (x, y, t, z)$$

to the original polygon. Divide finite columns by their positive z_i ; the original apex $(0, 0, 0, 1)$ becomes w_8 without a negative rescaling. Every four-minor is multiplied by the same determinant sign and by positive factors. The products and sign equalities in the piercing test are consequently unchanged. Thus the nine blocked-ear disjunctions are inherited from the original polygon, including the three ears involving the apex. No Euclidean triangle with a vertex at infinity is being assumed without this homogeneous justification.

These nine conditions are at source indices 2183, ..., 2191 (zero-based). Their Boolean expression and gate translation are certified, but Lemma 5.1 and its link to these expressions remain outside the present Lean proof.

6 Diagram extraction and the Conway coefficient

There are twenty pairs of nonadjacent shadow edges. For such a pair (i, j) , their open segments cross precisely when

$$\begin{aligned} T(i, i+1, j)T(i, i+1, j+1) &< 0, \\ T(j, j+1, i)T(j, j+1, i+1) &< 0. \end{aligned}$$

These strict segment-intersection conditions have been proved in Lean using Cramer's rule. At a crossing put $D_{ij} = \det(u_i, u_j)$. The intersection parameters lie in $(0, 1)$ and

$$\operatorname{sgn} D_{ij} = \operatorname{sgn} T(i, i+1, j+1), \quad D_{ji} = -D_{ij}.$$

For two finite chain edges the height of edge i minus that of edge j is H_{ij}/D_{ij} . Adopt the crossing-sign convention in which an overpassing first edge contributes $\operatorname{sgn} D_{ij}$. The crossing sign is therefore $\operatorname{sgn} H_{ij}$ and the first visit is under precisely when this differs from $\operatorname{sgn} D_{ij}$. For a crossing with edge seven, the first edge is always under; its crossing sign is $-\operatorname{sgn} D_{i7}$. Reversing all crossing signs would not affect the pair products below, but the code uses this single convention consistently.

For two live crossings on edge i , solving the line intersections gives

$$t_{ij} - t_{ik} = -\frac{C(i, j, k)}{D_{ij}D_{ik}}. \quad (7)$$

The nonconcurrency hypothesis makes the difference nonzero. Along a fixed edge, the earlier event is thus determined by the sign of $C(i, j, k)D_{ij}D_{ik}$. Events on different edges follow their cyclic edge order, with the basepoint at q_0 . This determines the complete Gauss traversal from the sign assignment. The formula explicitly includes twenty conditional direction relations and eighty conditional three-event transitivity relations. Their geometric validity, including the live-crossing guards, is proved in `DirectionRoots.lean` and `OrderRoots.lean`.

Write $\nabla_K(z) = 1 + a_2(K)z^2 + \dots$ for the Conway polynomial, normalized by $\nabla_{\text{unknot}} = 1$ and $\nabla_{K_+} - \nabla_{K_-} = z\nabla_{L_0}$. For a crossing c , let $\epsilon_c \in \{-1, 1\}$ be its sign and let U_c record whether its first visit in the based traversal is under. Two crossings are interleaved if exactly one endpoint of one Gauss chord lies between the endpoints of the other.

Lemma 6.1 (Based pair formula). *For any regular based knot diagram,*

$$2a_2(K) = \sum_{\{c,d\} \text{ interleaved, } U_c \neq U_d} \epsilon_c \epsilon_d. \quad (8)$$

Proof. Switch the crossings with $U_c = 1$, in the order of their first visits. The final diagram is descending and represents the unknot. The skein relation says that the change at c is ϵ_c times the linking number of its oriented smoothing. Twice that linking number is the sum of the current signs of crossings between the two smoothing components. These are exactly the chords interleaved with c .

Consider one interleaved unordered pair. If neither crossing is switched there is no contribution. If only one is switched, its contribution to twice the total change is the original product $\epsilon_c \epsilon_d$. If both are switched, the contribution when the first is switched uses the original sign of the second; when the second is switched it uses the reversed sign of the first. The two cancel. Summing and using $a_2(\text{unknot}) = 0$ proves the formula. In particular the formula does not require a crossing-minimal diagram. $\square$

The knot 8_1 has crossing number eight and Conway polynomial $1 - 3z^2$ [5]. The latter value is also recovered exactly from the identified ten-gon diagram in Section 8. Consequently its roof diagram, or a mirror diagram, satisfies

$$\sum_c [c \text{ is live}] \geq 8, \quad \sum_{\{c,d\} \text{ interleaved, } U_c \neq U_d} \epsilon_c \epsilon_d = -6.$$

These are the last two source constraints. The implementation tests interleaving by comparing the two visits of one crossing with the open interval between the visits of the other, uses inequality of the first-under flags, and adds +1 or −1 according to equality of the crossing signs. Inactive crossing pairs contribute zero. Thus its pseudo-Boolean sums are exactly the quantities displayed above.

The counter circuitry implementing these sums is proved correct in Lean for arbitrary boundary values. The interpretation as a_2 and the knot-theoretic lower bound on diagram crossings are external mathematical inputs; they are not consequences merely of counter correctness.

7 The finite exclusion

The base call `roof_boolean.build()` constructs the conjunction in Table 1. All optional strengthening flags are false: there is no canonical labelling, chord condition, Fox-colouring condition, or additional endpoint simplification in the certified formula.

Source family	Start	Count	Geometric Lean link
Point Pluecker	0	1260	proved
Line Pluecker	1260	280	proved
Line brackets	1540	48	proved
Homogeneous acyclicity	1588	126	proved
Projected acyclicity	1714	70	proved
Normalization	1784	1	proved
Line incidence	1785	48	proved
Four-line heights	1833	35	proved
Five-line heights	1868	315	proved
Blocked ears	2183	9	external
Crossing directions	2192	20	proved
Event transitivity	2212	80	proved
Conway target	2292	1	external
Crossing target	2293	1	external

Table 1: Zero-based source-root slices. “External” concerns geometric interpretation; every row has a certified Boolean-to-CNF translation.

The source is a 23,565-node acyclic expression graph. Its 23,563 ordinary Boolean nodes and two pseudo-Boolean nodes are connected to an explicit circuit of 18,454 gates. The gates are AND, Boolean equivalence, and majority; full adders and balanced counters are built from them. An additional constant variable gives 18,597 CNF variables in total. The roots include the 2,294 source conditions and the true constant. Each gate receives its defining equivalence clauses, and each root receives a unit clause. The resulting CNF has 76,298 clauses.

For a well-ordered gate list, an assignment to the inputs extends by evaluation to every gate. The defining clauses hold under this extension; root unit clauses hold exactly when their circuit outputs are true. Lean proves this fact for arbitrary such circuits. Local exhaustive checks connect each ordinary source operation to its gate implementation. For the actual counter circuits, bit-vector proofs establish equality and threshold semantics with sufficient

widths to exclude overflow. Signed weights are normalized using $-b = (1 - b) - 1$ for a Boolean value $b \in \{0, 1\}$. The actual gate-generated CNF is proved equal to the fixed CNF after semantics-preserving clause and literal normalization.

The fixed CNF was refuted by CaDiCaL. An independent DRAT checker produced an LRAT certificate, which was also accepted by a C LRAT checker. Clause identifiers were densely renumbered, preserving clauses and hints, for the Lean checker. The resulting proof has 1,577,701 actions. The fixed certificate and its input are available in the accompanying release; file identifiers and reproduction details are collected in Appendix A.

`FixedCertificate.lean` embeds the CNF and compressed proof bytes as fixed constants, decompresses and parses them through pure functions, and proves checker acceptance by `native_decide`. It then applies Lean’s standard `LRAT.check_sound` to obtain `Stick81.roofUnsat`. There is no acceptance hypothesis in this theorem.

Proposition 7.1 (Formal finite exclusion). *For the fixed source graph, no assignment satisfying its node semantics and its true constant makes all 2,294 source roots true. Moreover, for a positive nondegenerate normalized real frame, it suffices to assume the nine ear roots and the two target roots true to obtain a contradiction.*

Proof. The first assertion is `Stick81.source_formula_unsat`.

For the second, the algebraic, direction, and order lemmas supply 2,283 roots. The checked partition and eleven premises give all roots, so the first assertion applies. The resulting Lean theorem is named `geometric_exclusion_eleven` in namespace `Stick81.Submission`. $\square$

The geometric frame in this statement is actual real coordinate data, not an uninterpreted name for an arbitrary Boolean assignment. Its genericity, positive heights, and normalization are explicit hypotheses. The source-model hypothesis specifies evaluation of the acyclic source graph at the frame’s determinant signs. Such evaluation exists by recursion in source order; it is not a satisfiability assumption about its roots. In applying the proposition to a polygon, this recursive evaluation is the intended assignment.

8 The upper bound and proof of the main theorem

The file `known_10gon.tab` specifies ten triples of finite decimals. Every decimal is read as its exact rational value. The independent arithmetic checker `exact_polygon.py` examines all 35 pairs of nonadjacent edges and validates a regular projection with eighteen crossings, including intersection parameters and nonzero height differences. Adjacent edges are nondegenerate with distinct directions. These checks certify an embedded ten-gon, not its knot type alone.

The file `reidemeister_proposed.json` supplies thirty local moves. The standard-library checker `check_reidemeister.py` verifies each move on a four-valent rotation system with over/under parity: monogon or cancellable bigon removal for types I and II, and an empty triangular-face rewiring with consistent height order for type III. It checks continuity of the trace and a final orientation-preserving, over/under-preserving dart isomorphism to the PD code

```
[[1,9,2,8],[3,7,4,6],[5,12,6,13],[7,3,8,2],
 [9,1,10,16],[11,15,12,14],[13,4,14,5],[15,11,16,10]]
```

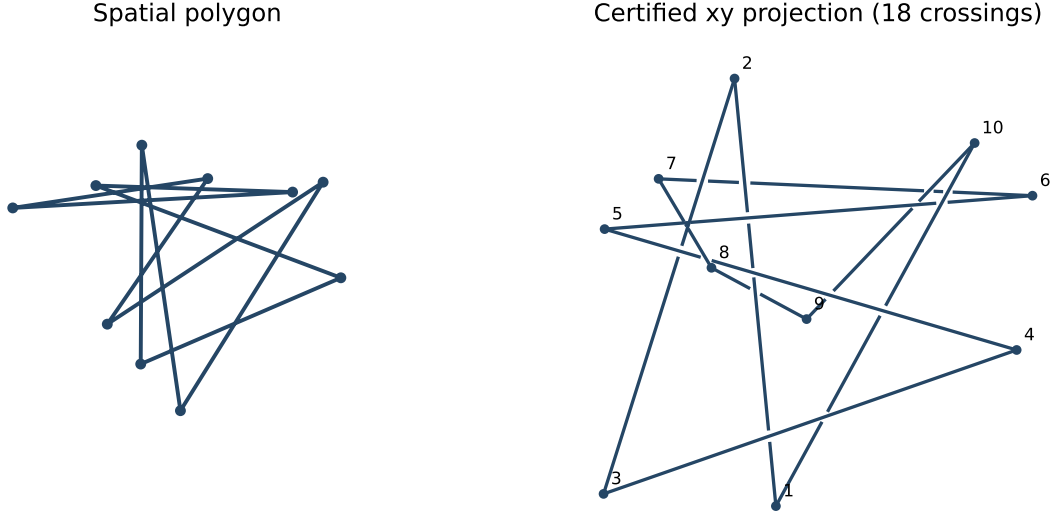


Figure 1: The rational ten-gon: a spatial illustration (left) and its certified projection (right). Vertices in this figure are numbered one through ten in file order. Underpass gaps on the right come from the exact crossing certificate. Floating-point coordinates are used only to draw the figure, not to verify the polygon.

This is also matched to Spherogram’s named 8_1 table entry [3]. Thus identification is based on a diagram equivalence trace, not on equality of a polynomial. The upper-bound certificate is independent of the SAT refutation and remains outside Lean. The two-bridge theorem [2] supplies an independent upper bound.

Proof of Theorem 1.1. Suppose that a nine-edge representative exists. Section 2 gives a positive generic normalized frame and a roof diagram of the knot or its mirror. Sections 3–6 show that its determinant signs satisfy all source constraints: the ordinary geometric lemmas supply the nine ears and two targets, and the real algebraic conditions supply the other 2,283. Evaluate the source graph at these signs and apply Proposition 7.1 to obtain a contradiction. Thus no nine-edge representative exists. Calvo’s classification excludes all smaller edge counts, and either upper-bound argument above gives ten edges. Therefore the minimum is ten. $\square$

9 Scope of the formalization

The formal development uses Lean 4.19.0 and a pinned version of mathlib; the exact versions are recorded in Appendix A. The audited final theorem depends on `propext`, `Classical.choice`, `Quot.sound`, and `Lean.ofReduceBool`. The last is the native-reflection trust boundary: the Lean compiler and runtime used for native computation are trusted in addition to the ordinary proof-checking machinery. There is no `sorryAx` in this theorem’s audited dependency list and no new axiom asserting the knot theorem or the eleven missing geometric implications. Solver heuristics and the compression program are not trusted to establish unsatisfiability: their output must pass the certified checker.

The formal development establishes the geometric implications for 2,283 of the 2,294 source constraints. The global coverage argument, the nine ear conditions, and the two diagram-invariant conditions are supplied by the ordinary mathematical proofs in Sections 2, 5, and 6. These counts describe the scope of formalization, not a probability of correctness. The lower

bound depends on every necessary condition and on the global reduction.

Exact rational controls, an independent Alexander-coefficient calculation, and separate certificate replays provide supplementary checks; they do not establish universal geometric coverage. The development does not yet contain an end-to-end theorem `stick(K81)=10`.

The code, certificates, and verification records are available in the fixed release cited in Appendix A. They permit reproduction of the finite checks separately from review of the geometric and topological arguments.

Use of generative AI. Generative AI tools, including OpenAI Codex, were used extensively in developing and reviewing mathematical arguments, implementing the computational and Lean components, and drafting this manuscript. The scope and limitations of formal verification are stated explicitly. The author assumes responsibility for the submitted work, including its mathematical claims, code, and references. AI-assisted review is not independent human peer review.

License. The manuscript and original figures are licensed under CC BY 4.0. The original accompanying code is available under the MIT License; third-party components retain their existing licenses.

A Computational data and reproduction

The source repository is

<https://github.com/griffithwang/stick8-1-certificate>.

The verified release `v0.1.0-audit` is pinned to commit

`0b88e7c65ab01fbe281920946d58ae079191ec39`

It contains the source dependency closure, finite data, and a hash manifest. The large certificates are distributed as release assets. The file sizes below refer to the exact distributed bytes.

File	Bytes	Role
<code>roof_direct.cnf</code>	1,464,784	Fixed CNF instance
<code>roof_base_dense.lrat</code>	700,460,899	Dense textual LRAT proof
<code>roof.lz4.hex</code>	408,064,310	Compressed proof for Lean

The last file is installed in `lean_formalization/`. Their SHA-256 checksums identify the files used in verification; a matching checksum establishes file consistency, not mathematical validity.

`roof_direct.cnf`
`b7034c3c955b2cf9c3a7878439f25941c0f51c5eeebb421b727d2ebfd2227f7c`

`roof_base_dense.lrat`
`97d9c4224e353ea24ac09b5373be1fb217d2ef057577cb6f01b6475b05190a87`

`lean_formalization/roof.lz4.hex`
`3185a627790f0c17bdc4773990f17d55716d003b6ef8ca903c50f0031b75d3de`

Software versions and checks. Lean is pinned to version 4.19.0 and mathlib to commit c44e0c8ee63ca166450922a373c7409c5d26b00b

All 59 local Lean modules were rebuilt successfully on Windows using the pinned upstream mathlib cache. Per-module source hashes, exit codes, and timings are recorded in `LEAN_RELEASE_VERIFICATION.json`. The independent arithmetic and C-checker replays have separate reports; no claim is made that every platform was tested.

After installing the pinned dependencies and obtaining the release assets as described in the repository README, run from the repository root:

```
python verify_public.py
python verify_partial_formalization.py --full
```

The first command checks the file manifest, replays the exact upper-bound trace, and reconstructs and checks the base encoding. The second recompiles all local Lean modules. Omitting `--full` performs only the incremental final-theorem audit. The arithmetic checkers use assertions, so Python must be run without its optimization flag.

An independent LRAT replay on a system with a C compiler is:

```
cc -O2 lrat-check.c -o lrat-check
./lrat-check roof_direct.cnf roof_base_dense.lrat
```

References

- [1] J. A. Calvo, *Geometric Knot Spaces and Polygonal Isotopy*, arXiv:math/9904037v2 (1999), Theorem 1(iv).
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